Test note

June 14, 2026

Abstract

This is a test note for embedding a pdf to my homepage.

1 Standard cosmology

Assuming our universe is a homogeneous and isotropic spacetime, the metric tensor is given by

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -dt^2 + a^2(t)\left(dx^2 + dy^2 + dz^2\right), \tag{1}$$

or explicitly

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2(t) & 0 & 0 \\ 0 & 0 & a^2(t) & 0 \\ 0 & 0 & 0 & a^2(t) \end{pmatrix}, \tag{2}$$

where ds^2 is line element, x, y, z are flat-spatial coordinates, t is cosmic time, and a is scale factor. $g_{\mu\nu}$ is called the Friedmann-Lemaître-Robertson-Walker (FLRW) metric.

According to general relativity, the relation between spacetime geometry and the distribution of all components in our universe is formulated by the Einstein equation

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{3}$$

where G is the gravitational constant, $T_{\mu\nu}$ is energy momentum tensor, and the Einstein tensor $G_{\mu\nu}$ is defined with Ricci tensor $R_{\mu\nu}$ and Ricci scalar R by

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R. \tag{4}$$

We consider the Einstein equation under the FLRW metric. When assuming that our universe is isotropic and homogeneous, we can determine the energy momentum tensor explicitly,

$$T^\mu{}_\nu = \begin{pmatrix} -\rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix} \tag{5}$$

where ρ is the energy density of the universe and P is the pressure. Substituting FLRW and the above $T^\mu{}_\nu$ to the Einstein equation, we get two equations, called Friedmann equations,

$$\left(\frac{1}{a}\frac{da}{dt}\right)^2 = \frac{8\pi G}{3}\rho \tag{6}$$

$$\frac{1}{a}\frac{d^2 a}{dt^2} = -\frac{4\pi G}{3}\left(\rho + 3P\right) \tag{7}$$